

Practical 2: Working with Data

Data Science — Academic Lab Report

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Abstract

This report documents the implementation and experimental evaluation of Practical 2 (*Working with data*) in R. The study covers end-to-end data manipulation procedures across standard statistical datasets (`titanic.csv`, `cars.csv`, `airquality.csv`, `sales.txt`, and relational customer data). Key techniques analyzed include exploratory data inspection, graphic customization, dataset mutation and sorting, conditional data imputation and discretization (equal-width vs. quantile binning), date arithmetic, frequency de-aggregation, correlation analysis, simple and stratified sampling, multi-level aggregation, and relational table joining.

1 Exercise 1: Inspection of Data

1.1 Objective and Commands

The `titanic.csv` dataset contains historical survival records from the Titanic disaster. The dataset is ingested using `read.csv()`, its schema is inspected, the redundant indexing column `X` is pruned using `subset()`, and variable properties are evaluated through structural inspection.

```
# Ingestion and redundant index removal
titanic <- read.csv("titanic.csv", header = TRUE, sep = ",")
names(titanic)
titanic <- subset(titanic, select = -X)

# Structural and statistical inspection
head(titanic)
summary(titanic)
str(titanic)
```

1.2 Outputs and Console Verification

```
> names(titanic)
[1] "X"          "Class"      "Sex"        "Age"        "Survived"  "Freq"
```

```
> head(titanic)
  Class  Sex Age Survived Freq
1  1st Male Child      No    0
2  2nd Male Child      No    0
3  3rd Male Child      No   35
4 Crew Male Child      No    0
5  1st Female Child     No    0
```

```
6    2nd Female Child          No      0
```

```
> summary(titanic)
      Class                Sex                Age                Survived
Length:32      Length:32      Length:32      Length:32
Class :character Class :character Class :character Class :character
Mode  :character Mode  :character Mode  :character Mode  :character
      Freq
Min.   : 0.00
1st Qu.: 0.75
Median : 13.50
Mean   : 68.78
3rd Qu.: 77.00
Max.   :670.00
```

1.3 Discussion: Variable Types Identification

- **Quantitative Variables:** The only quantitative variable in the dataset is `Freq` (integer). It represents the absolute count of individuals sharing a specific demographic profile. In `summary()`, quantitative variables are identified by the presence of continuous statistical metrics (minimum, median, mean, quartiles, and maximum). In `str()`, it is typed as `int`.
- **Categorical Variables:** `Class` (1st, 2nd, 3rd, Crew), `Sex` (Male, Female), `Age` (Child, Adult), and `Survived` (No, Yes) are nominal/ordinal categorical attributes. They represent discrete non-continuous modalities and are typed as `character` or `factor` without summary distribution statistics.

2 Exercise 2: Working with Basic Graphics

2.1 Objective and Commands

The `cars.csv` dataset records automobile speed (`speed` in mph) and stopping distance (`dist` in ft). Basic exploratory visualizations (scatter plots and frequency histograms) are constructed and stylized with distinct color palettes, informative axis labels, and exported directly to PDF vector format.

```
cars <- read.csv("cars.csv", header = TRUE, sep = ",")
if ("X" %in% colnames(cars)) cars <- subset(cars, select = -X)

# Customized Graphics exported to PDF
pdf("plot_cars_scatter.pdf")
plot(cars$speed, cars$dist,
      main = "Stopping Distance vs Speed",
      xlab = "Speed (mph)", ylab = "Stopping Distance (ft)",
      col = "darkblue", pch = 19)
dev.off()

pdf("plot_cars_hist_dist.pdf")
hist(cars$dist, main = "Distribution of Stopping Distance",
      xlab = "Stopping Distance (ft)", ylab = "Frequency",
      col = "steelblue", border = "white")
dev.off()

pdf("plot_cars_hist_speed.pdf")
hist(cars$speed, main = "Distribution of Speed",
      xlab = "Speed (mph)", ylab = "Frequency",
      col = "coral", border = "white")
dev.off()
```

2.2 Visual Inspection

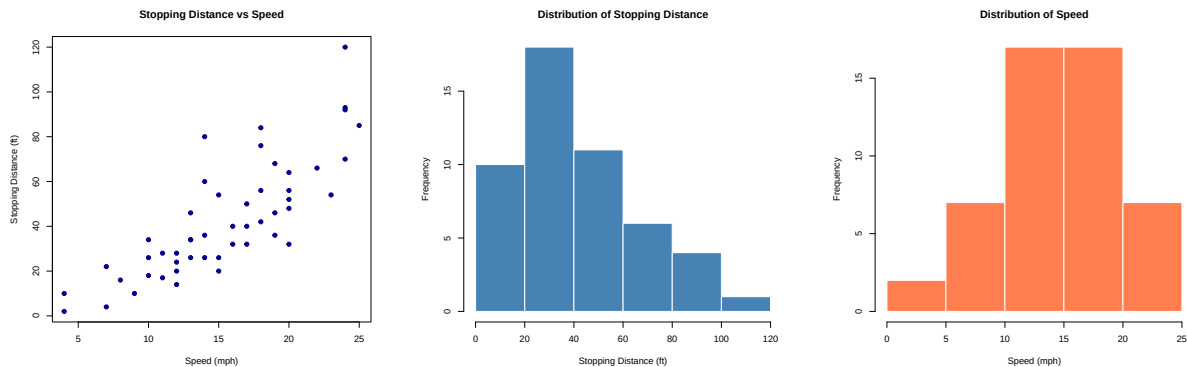


Figure 1: Distance vs. Speed Figure 2: Distance Histogram Figure 3: Speed Histogram

The scatter plot highlights an upward curvilinear association between travel velocity and braking displacement. The distance histogram reveals a right-skewed distribution, whereas automobile speed displays an approximately symmetric profile.

3 Exercise 3: Transformations of Variables and Datasets

3.1 Objective and Commands

The base `cars` data frame is augmented with two newly acquired observations: (`speed = 21, dist = 47`) and (`speed = 34, dist = 87`). The combined data frame is subsequently ordered along the `speed` dimension using direct array indexing with `order()` and contextual scope evaluation with `with()`.

```
# Construct auxiliary data frame
new_cars <- data.frame(speed = c(21, 34), dist = c(47, 87))

# Row-bind datasets
cars_extended <- rbind(cars, new_cars)

# Sort ascending by speed
cars_sorted1 <- cars_extended[order(cars_extended$speed), ]
cars_sorted2 <- cars_extended[with(cars_extended, order(speed)), ]
identical(cars_sorted1, cars_sorted2)
```

3.2 Outputs

```
> tail(cars_extended)
  speed dist
47    24   92
48    24   93
49    24  120
50    25   85
51    21   47
52    34   87

> identical(cars_sorted1, cars_sorted2)
[1] TRUE

> tail(cars_sorted1, 10)
  speed dist
47    24   92
48    24   93
49    24  120
50    25   85
51    21   47
52    34   87
```

43	20	64
51	21	47
44	22	66
45	23	54
46	24	70
47	24	92
48	24	93
49	24	120
50	25	85
52	34	87

Both sorting methods achieve identical permutations. In the final sorted structure, observation 51 (speed = 21) is positioned between records with speeds 20 and 22, while observation 52 (speed = 34) represents the maximum boundary.

4 Exercise 4: Data Manipulation

4.1 Objective and Commands

Exploratory slicing, missing-value profiling, and conditional aggregation are performed on `airquality.csv`.

```
air <- read.csv("airquality.csv", header = TRUE, sep = ",")
if ("X" %in% colnames(air)) air <- subset(air, select = -X)

# 1. First 2 rows
head(air, 2)

# 2. Number of observations
nrow(air)

# 3. Ozone value at row 40
air[40, "Ozone"]

# 4. Total missing values in Ozone
sum(is.na(air$Ozone))

# 5. Ozone sample mean excluding NA
mean(air$Ozone, na.rm = TRUE)

# 6. Subset: Ozone > 31 & Temp > 90; compute Solar.R mean
subset_air <- subset(air, Ozone > 31 & Temp > 90)
mean(subset_air$Solar.R, na.rm = TRUE)
```

4.2 Console Results

```
> head(air, 2)
  Ozone Solar.R Wind Temp Month Day
1   41    190   7.4   67     5    1
2   36    118   8.0   72     5    2

> nrow(air)
[1] 153

> air[40, "Ozone"]
[1] 71

> sum(is.na(air$Ozone))
[1] 37

> mean(air$Ozone, na.rm = TRUE)
```

```
[1] 42.12931

> mean(subset_air$Solar.R, na.rm = TRUE)
[1] 212.8
```

- Row 40 reports an Ozone concentration of **71** ppb.
- Exactly **37** records out of 153 are unobserved (NA) in the **Ozone** column (24.18%).
- The unweighted mean Ozone level is $\bar{x} \approx 42.13$ ppb.
- When conditioning on high-temperature, high-ozone episodes ($\text{Ozone} > 31 \wedge \text{Temp} > 90$), mean solar radiation is $\bar{x}_{\text{Solar.R}} = 212.8$ langleys.

5 Exercise 5: Data Transformation (Discretization and Dates)

5.1 Objective and Commands

Continuous features are discretized using two core strategies: equal-width partitioning via `cut()` and equal-frequency (quantile) binning via sample quantiles, explicitly allocating missing entries to a dedicated level (`binNA`). A sequential elapsed-time metric `AbsDay` is constructed from calendar attributes.

```
# 5.1 Discretise Ozone into 5 equal-width bins + binNA
ozone_cut <- cut(air$Ozone, breaks = 5, labels = paste0("bin", 1:5),
  include.lowest = TRUE)
air$Ozone_bin <- as.character(ozone_cut)
air$Ozone_bin[is.na(air$Ozone_bin)] <- "binNA"
air$Ozone_bin <- factor(air$Ozone_bin, levels = c(paste0("bin", 1:5), "binNA"))

# 5.2 Discretise Solar.R into 4 equal-size (quantile) bins + binNA
q_solar <- quantile(air$Solar.R, probs = seq(0, 1, 0.25), na.rm = TRUE)
solar_cut <- cut(air$Solar.R, breaks = q_solar, labels = paste0("bin", 1:4),
  include.lowest = TRUE)
air$Solar_bin <- as.character(solar_cut)
air$Solar_bin[is.na(air$Solar_bin)] <- "binNA"
air$Solar_bin <- factor(air$Solar_bin, levels = c(paste0("bin", 1:4), "binNA"))

# 5.3 Elapsed calendar days from reference date (1973-05-01)
current_dates <- as.Date(paste(1973, air$Month, air$Day, sep = "-"))
air$AbsDay <- as.numeric(current_dates - as.Date("1973-05-01"))
```

5.2 Outputs and Empirical Distributions

```
> table(air$Ozone_bin)
 bin1  bin2  bin3  bin4  bin5 binNA
   62   29   18    5    2   37

> table(air$Solar_bin)
 bin1  bin2  bin3  bin4 binNA
   37   36   36   37    7
```

- **Equal-width Partitioning (Ozone):** Divides the range into uniform intervals. Because ozone values exhibit positive skewness, the majority of observations are clustered in `bin1` ($n = 62$) and `bin2` ($n = 29$), while upper bins are sparsely populated.

- **Equal-size Partitioning (Solar.R):** Quantile boundaries guarantee an equitable observation balance (≈ 36 – 37 records per active bin) alongside the 7 unrecorded instances in `binNA`.
- **Elapsed Day Count (AbsDay):** Spans values from 0 (May 1st) to 152 (September 30th), mapping 153 consecutive temporal indices across the summer monitoring campaign.

6 Exercise 6: Data Transformation (Titanic Frequency De-aggregation)

6.1 Objective and Commands

Ordinal categorization is mapped to numerical ranks (3rd = 1, 2nd = 2, 1st = 3, Crew = 4). The aggregated contingency table is expanded into an individual-level data frame (`titanic2`) by replicating rows according to `Freq`.

```
# Numerise class attribute
class_mapping <- c("3rd" = 1, "2nd" = 2, "1st" = 3, "Crew" = 4)
titanic$Class_num <- class_mapping[as.character(titanic$Class)]

# De-aggregate contingency data frame using rep() and Freq
titanic2 <- titanic[rep(seq_len(nrow(titanic)), titanic$Freq), ]
titanic2 <- subset(titanic2, select = -Freq)
nrow(titanic2)

# Graphical comparison
pdf("plot_titanic_original.pdf", width = 7, height = 5)
plot(titanic, main = "Original Titanic (Aggregated Structure)")
dev.off()

pdf("plot_titanic2_individual.pdf", width = 7, height = 5)
plot(as.data.frame(unclass(titanic2), stringsAsFactors = TRUE),
     main = "De-aggregated Titanic (Individual Records)")
dev.off()
```

6.2 Outputs and Comparative Discussion

```
> table(titanic$Class, titanic$Class_num)
      1 2 3 4
1st   0 0 8 0
2nd   0 8 0 0
3rd   8 0 0 0
Crew  0 0 0 8

> nrow(titanic2)
[1] 2201
```

The original data frame comprised 32 cross-tabulated contingency records, weighting each profile equally on standard scatter/mosaic plots regardless of frequency. The reconstructed data frame expands into exactly $N = 2201$ passenger records. The corresponding mosaic plot accurately displays real demographic imbalances (e.g., substantial crew and 3rd class representation, male majority, and differential survival ratios).

7 Exercise 7: Data Selection and Sampling

7.1 Correlation Matrices

Pearson correlation matrices are evaluated using complete pairwise observations.

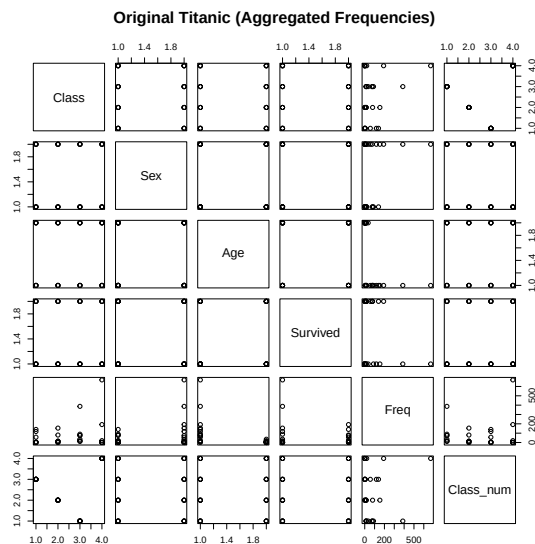


Figure 4: Aggregated Data Plot

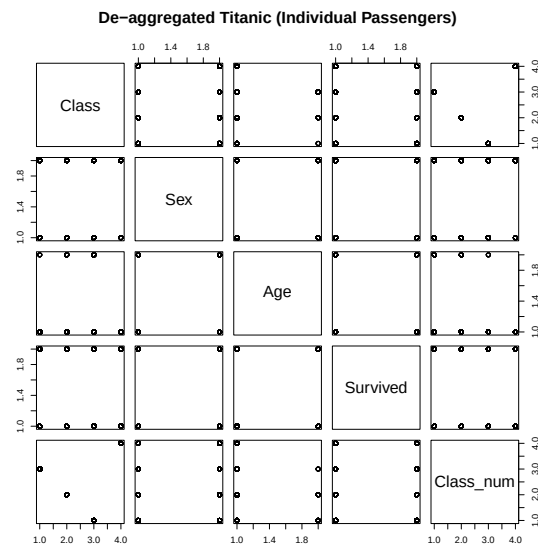


Figure 5: Individual Passenger Mosaic

```
> round(cor_air, 3)
      Ozone Solar.R   Wind   Temp  Month   Day
Ozone   1.000   0.348 -0.612  0.699  0.143 -0.005
Solar.R  0.348   1.000 -0.127  0.294 -0.074 -0.058
Wind    -0.612  -0.127  1.000 -0.497 -0.194  0.050
Temp     0.699   0.294 -0.497  1.000  0.404 -0.097
Month    0.143  -0.074 -0.194  0.404  1.000 -0.009
Day     -0.005  -0.058  0.050 -0.097 -0.009  1.000

> round(cor_cars, 3)
      speed dist
speed  1.000 0.807
dist   0.807 1.000
```

- **Redundancy in air:** No pair of attributes satisfies strict redundancy ($|r| \approx 1$). Moderate collinearity exists between **Temp** and **Ozone** ($r = 0.699$) and inverse correlation between **Wind** and **Ozone** ($r = -0.612$). All variables retain distinct informational value.
- **Redundancy in cars:** A high positive correlation is observed ($r = 0.807$). While stopping distance physically depends on kinetic energy ($\propto v^2$), both attributes measure separate mechanical properties. However, in low-complexity predictive models, this suggests substantial mutual information.

7.2 Sampling Strategies

```
# Simple random sampling (n = 50)
set.seed(123)
sample_simple <- air[sample(1:nrow(air), size = 50, replace = FALSE), ]
nrow(sample_simple)

# Stratified sampling (5 observations per month)
stratified_idx <- unlist(lapply(split(1:nrow(air), air$Month), function(idx) {
  sample(idx, size = 5, replace = FALSE)
}))
sample_stratified <- air[stratified_idx, ]
table(sample_stratified$Month)
```

```
> nrow(sample_simple)
[1] 50

> table(sample_stratified$Month)
5 6 7 8 9
5 5 5 5 5
```

Simple random sampling yields $n = 50$ instances drawn uniformly at random. Stratified sampling enforces month-level quota constraints, guaranteeing balanced seasonal representations ($n = 25$ total).

8 Exercise 8: Data Aggregation

8.1 Objective and Commands

The commercial transactions stored in `sales-1.txt` are summarized using the `aggregate()` function across store entities and product categories.

```
sales <- read.table("sales-1.txt", header = TRUE, stringsAsFactors = FALSE)

# Total sales per store
total_sales_store <- aggregate(sales_amount ~ store, data = sales, FUN = sum)

# Average sales per product category
avg_sales_cat <- aggregate(sales_amount ~ product_category, data = sales, FUN =
  mean)

# Multi-factor aggregation: store and category
total_store_cat <- aggregate(sales_amount ~ store + product_category, data =
  sales, FUN = sum)

# Bar chart generation and export
pdf("plot_sales_per_store.pdf", width = 6, height = 4.5)
barplot(total_sales_store$sales_amount,
  names.arg = total_sales_store$store,
  col = "steelblue", border = "black",
  main = "Total Sales per Store",
  xlab = "Store Entity", ylab = "Cumulative Sales Amount ($)")
dev.off()
```

8.2 Aggregated Results

Table 1: Summarized Sales Metrics Across Stores and Product Categories

Aggregation Dimension	Level / Group	Metric Value
Total Sales per Store	Store_A	\$2,430.00
	Store_B	\$2,980.00
	Store_C	\$1,820.00
Mean Sales per Category	Clothing	\$660.00
	Electronics	\$1,463.33
	Groceries	\$430.00

Store_B generated the highest aggregate revenue (\$2,980), followed by Store_A (\$2,430) and Store_C (\$1,820). Across merchandise types, Electronics exhibits the highest mean transaction value (\$1,463.33).

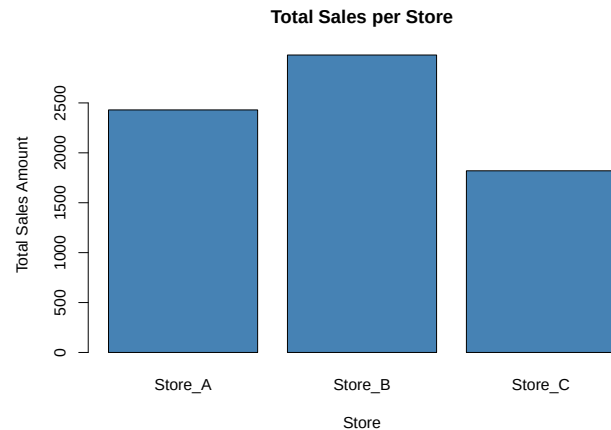


Figure 6: Cumulative sales performance broken down by individual retail store.

9 Exercise 9: Merging Relational Datasets

9.1 Objective and Commands

Relational joining is evaluated by matching customer records from `customers.txt` against transaction orders in `orders.txt` along the primary/foreign key `customer_id`.

```
# Ingestion
customers <- read.table("customers.txt", header = TRUE, stringsAsFactors =
  FALSE)
orders <- read.table("orders.txt", header = TRUE, stringsAsFactors = FALSE)

# Inner join across customer_id
customer_orders <- merge(customers, orders, by = "customer_id", all = FALSE)

# Count of unique matched customers
unique_cust_count <- length(unique(customer_orders$customer_id))

# Total order frequency per customer
orders_per_customer <- table(customer_orders$customer_id)

# Export merged data frame
write.csv(customer_orders, file = "customer_orders.csv", row.names = FALSE)
```

9.2 Discussion

The inner join operation filters records to preserve only those customer identifiers present simultaneously in both tables. Customer purchase histories are aggregated, quantifying repeat-buyer engagement metrics prior to persisting the consolidated relational view into `customer_orders.csv`.

Appendix: Complete Execution Log

The following block contains the verbatim console output generated from running the complete script `pract2.R` sequentially:

```
[1] "X"      "Class"  "Sex"    "Age"    "Survived" "Freq"
      Class   Sex   Age Survived Freq
1   1st  Male Child     No     0
2   2nd  Male Child     No     0
3   3rd  Male Child     No    35
4   Crew Male Child     No     0
5   1st Female Child     No     0
6   2nd Female Child     No     0
7   3rd Female Child     No    17
8   Crew Female Child     No     0
9   1st  Male Adult     No   118
10  2nd  Male Adult     No  154
11  3rd  Male Adult     No  387
12  Crew Male Adult     No  670
13  1st Female Adult     No     4
14  2nd Female Adult     No    13
15  3rd Female Adult     No    89
16  Crew Female Adult     No     3
17  1st  Male Child     Yes     5
18  2nd  Male Child     Yes    11
19  3rd  Male Child     Yes    13
20  Crew Male Child     Yes     0
21  1st Female Child     Yes     1
22  2nd Female Child     Yes    13
23  3rd Female Child     Yes    14
24  Crew Female Child     Yes     0
25  1st  Male Adult     Yes    57
26  2nd  Male Adult     Yes    14
27  3rd  Male Adult     Yes    75
28  Crew Male Adult     Yes   192
29  1st Female Adult     Yes   140
30  2nd Female Adult     Yes    80
31  3rd Female Adult     Yes    76
32  Crew Female Adult     Yes    20
      Class   Sex   Age Survived Freq
1   1st  Male Child     No     0
2   2nd  Male Child     No     0
3   3rd  Male Child     No    35
4   Crew Male Child     No     0
5   1st Female Child     No     0
6   2nd Female Child     No     0
      Class   Sex   Age Survived
Length:32      Length:32      Length:32      Length:32
Class :character Class :character Class :character Class :character
Mode  :character Mode  :character Mode  :character Mode  :character

      Freq
Min.   : 0.00
1st Qu.: 0.75
Median : 13.50
Mean   : 68.78
3rd Qu.: 77.00
Max.   :670.00
'data.frame': 32 obs. of 5 variables:
 $ Class : chr "1st" "2nd" "3rd" "Crew" ...
 $ Sex   : chr "Male" "Male" "Male" "Male" ...
 $ Age   : chr "Child" "Child" "Child" "Child" ...
 $ Survived: chr "No" "No" "No" "No" ...
 $ Freq   : int 0 0 35 0 0 0 17 0 118 154 ...
X speed dist
1 1 4 2
2 2 4 10
3 3 7 4
4 4 7 22
5 5 8 16
6 6 9 10
pdf
2
pdf
2
pdf
2
speed dist
1 4 2
2 4 10
3 7 4
4 7 22
5 8 16
6 9 10
speed dist
1 21 47
2 34 87
speed dist
47 24 92
48 24 93
49 24 120
50 25 85
51 21 47
52 34 87
[1] TRUE
```

```

    speed dist
43    20    64
51    21    47
44    22    66
45    23    54
46    24    70
47    24    92
48    24    93
49    24   120
50    25    85
52    34    87
  Ozone Solar.R Wind Temp Month Day
1     41     190  7.4   67     5     1
2     36     118  8.0   72     5     2
  Ozone Solar.R Wind Temp Month Day
1     41     190  7.4   67     5     1
2     36     118  8.0   72     5     2
[1] 153
[1] 71
[1] 37
[1] 42.12931
  Ozone Solar.R Wind Temp Month Day
69     97     267  6.3   92     7     8
70     97     272  5.7   92     7     9
120    76     203  9.7   97     8    28
121    118     225  2.3   94     8    29
122     84     237  6.3   96     8    30
123     85     188  6.3   94     8    31
124     96     167  6.9   91     9     1
125     78     197  5.1   92     9     2
126     73     183  2.8   93     9     3
127     91     189  4.6   93     9     4
[1] 212.8

bin1 bin2 bin3 bin4 bin5 binNA
  62   29   18    5    2   37

bin1 bin2 bin3 bin4 binNA
  37   36   36   37    7
  Month Day AbsDay Ozone_bin Solar_bin
1      5  1      0    bin2    bin2
2      5  2      1    bin2    bin2
3      5  3      2    bin1    bin2
4      5  4      3    bin1    bin4
5      5  5      4    binNA    binNA
6      5  6      5    bin1    binNA
  Month Day AbsDay Ozone_bin Solar_bin
148     9  25   147    bin1    bin1
149     9  26   148    bin1    bin2
150     9  27   149    binNA    bin2
151     9  28   150    bin1    bin2
152     9  29   151    bin1    bin2
153     9  30   152    bin1    bin3

    1 2 3 4
1st  0 0 8 0
2nd  0 8 0 0
3rd  8 0 0 0
Crew 0 0 0 8
[1] 2201
  Class Sex Age Survived Class_num
3     3rd Male Child      No      1
3.1   3rd Male Child      No      1
3.2   3rd Male Child      No      1
3.3   3rd Male Child      No      1
3.4   3rd Male Child      No      1
3.5   3rd Male Child      No      1
pdf
2
pdf
2
  Ozone Solar.R Wind Temp Month Day
Ozone  1.000  0.348 -0.612  0.699  0.143 -0.005
Solar.R 0.348  1.000 -0.127  0.294 -0.074 -0.058
Wind    -0.612 -0.127  1.000 -0.497 -0.194  0.050
Temp     0.699  0.294 -0.497  1.000  0.404 -0.097
Month    0.143 -0.074 -0.194  0.404  1.000 -0.009
Day     -0.005 -0.058  0.050 -0.097 -0.009  1.000
  speed dist
speed 1.000 0.807
dist  0.807 1.000
[1] 50
  Ozone Solar.R Wind Temp Month Day Ozone_bin Solar_bin AbsDay
14     14     274 10.9   68     5    14    bin1    bin4    13
50     12     120 11.5   73     6    19    bin1    bin2    49
118     73     215  8.0   86     8    26    bin3    bin3    117
43      NA     250  9.2   92     6    12    binNA    bin3    42
153     20     223 11.5   68     9    30    bin1    bin3    152
151     14     191 14.3   75     9    28    bin1    bin2    150

5 6 7 8 9
5 5 5 5 5
[1] 25
  store product_category sales_amount
1 Store_A Electronics      1200
2 Store_A Groceries        450
3 Store_A Clothing         780
4 Store_B Electronics     2300
5 Store_B Clothing         680

```

```
6 Store_C      Groceries      410
[1] "store"      "product_category" "sales_amount"
      store sales_amount
1 Store_A      2430
2 Store_B      2980
3 Store_C      1820
  product_category sales_amount
1      Clothing      660.000
2      Electronics  1463.333
3      Groceries    430.000
  store product_category sales_amount
1 Store_A      Clothing      780
2 Store_B      Clothing      680
3 Store_C      Clothing      520
4 Store_A      Electronics  1200
5 Store_B      Electronics  2300
6 Store_C      Electronics   890
7 Store_A      Groceries    450
8 Store_C      Groceries    410
pdf
2
```

Console Output: output-pract2.txt